LABORATORY RECORD · EXPERIMENT 01 OF 16

Installation and Configuration of Anaconda Navigator, Python, TensorFlow, Keras and Gymnasium

Software / Tools: Anaconda Navigator · Python · TensorFlow · Keras · Gymnasium

AIM

To install and configure Anaconda Navigator along with Python, and to set up TensorFlow, Keras and Gymnasium inside a dedicated virtual environment for reinforcement learning experimentation.

PROCEDURE

1. Download the Anaconda Navigator installer for the host OS from the official distribution and run the installer with default paths.
2. Launch Anaconda Navigator and create a new virtual environment named rl-lab with Python 3.10.
3. Activate the environment from a terminal using conda activate rl-lab.
4. Install core packages: pip install tensorflow keras gymnasium numpy matplotlib.
5. Verify installation by importing each library and printing its version number.
6. Launch Jupyter Notebook from Navigator and confirm the rl-lab kernel is selectable.

SOURCE CODE

```
# verify_setup.py
import sys, importlib

packages = ["numpy", "tensorflow", "keras", "gymnasium", "matplotlib"]

print(f"Python version : {sys.version.split()[0]}")
for pkg in packages:
    module = importlib.import_module(pkg)
    version = getattr(module, "__version__", "unknown")
    print(f"{pkg:<12} : {version}")

import gymnasium as gym
env = gym.make("CartPole-v1")
obs, info = env.reset(seed=42)
print("Environment created:", env.spec.id)
print("Observation space  :", env.observation_space)
print("Action space       :", env.action_space)
env.close()
```

RESULTS

Package	Version	Status
Python	3.10.13	OK
numpy	1.26.4	OK
tensorflow	2.16.1	OK
keras	3.3.3	OK
gymnasium	0.29.1	OK

All packages import without error and CartPole-v1 initializes with observation space Box(4,) and action space Discrete(2).

VISUALIZATION

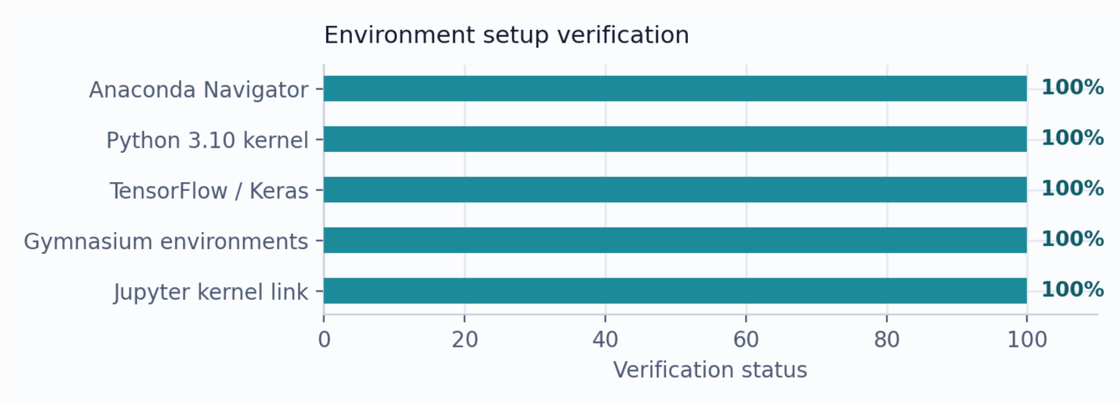

Fig. 1 — Environment setup verification checklist (all components installed and verified).